

ADR-009: Orchestration Evaluation Substrate (Derive-Only)

FR017 — [sources/adr.md](#)

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	Accepted (FR-017 M1–M4 close-out — **Complete / Frozen**)
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

Contents

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Status: Accepted (FR-017 M1–M4 close-out — **Complete / Frozen**)

Date: 2026-08-07

Acceptance: [eval/fr017_agent_evaluation_observability.md](#)

M4: [eval/fr017_m4_evaluation.md](#)

Contracts: [eval/fr017_m1_observability_contracts.md](#)

Spike: [eval/fr017_m0_engineering_spike.md](#) (Accepted — narrow GO)

Reaffirms: [ADR-007](#) (BOPA metrics ownership); [ADR-008](#) (FR-016 audits are learning-proof SoT for orchestration authority; daily prep remains `cic agent run`); [ADR-003](#) (FR-008 owns workflow traces); [ADR-006](#) (truth engine not duplicated)

Does not amend: ADR-001, ADR-002, ADR-004, ADR-005.

Context

FR-017's historical laundry list largely duplicated frozen FR-008 / FR-014 / FR-015 / FR-016 work. M0 rejected dashboards and framework observability and recommended a **narrow derive-only** evaluation substrate over existing orchestration audits, with reconstructability R1–R12 as first-class criteria. Horizon 1B must not be blocked on FR-017.

Owner accepted M0 and authorised M1 contracts only, with an explicit ban on DOS / BOPA / OBS / Handoff / AgentRun / orchestration runtime changes unless a proven contract blocker is returned for approval.

Decision

1. **Introduce** `career_intelligence.multi_agent.observability` as a **pure, derive-only** module: `OrchestrationRunMetrics`, handoff metrics, corpus aggregates, parent/child correlation, and `evaluate_reconstructability` (R1–R12).
2. **Reuse** FR-015 `AgentRunMetrics` for child roll-ups. Callers may pass child metrics; FR-017 must not redefine BOPA observability or invent a second AgentRun metrics schema.
3. **Preserve missing vs zero:** count emptiness as `0`; optional provider/model/token/cost/latency absence as `None` — never coerce missing to zero.
4. **Do not** add dashboards, telemetry stores, tracing platforms, or new audit event kinds. Thin read-only metrics CLI and corpus suite are in scope for M2–M3; M4 freezes without broadening.
5. **Do not** modify DOS, BOPA, OBS, Handoff, AgentRun, DelegationPolicy, or ToolPolicy unless a genuine R1–R12 contract blocker is proven and owner-approved.

6. **Horizon 1B coupling:** FR-017 must **not** gate FR-018+. Usable application loop (FR-008–FR-015) remains the 1B engineering gate.
 7. **Product posture:** learning / substrate. Prefer `cic agent run` for ordinary preparation; orchestration evaluation does not become the daily default.
 8. **Freeze (M4):** FR-017 is Complete / Frozen under [eval/fr017_agent_evaluation_observability.md](#). Do not reopen without owner request.
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Consequences

- Evaluation becomes **derived views over audits**, consistent with FR-015 M4.
 - Reconstructability is testable without live LLM or new instrumentation.
 - Scope pressure toward dashboards or runtime instrumentation is rejected by this ADR unless owner revisits.
 - M2 may package FR-016 corpus cases against these contracts without changing specialist behaviour.
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Alternatives considered

Alternative	Disposition
Full laundry-list / dashboard product	Rejected (M0)
New DOS event kinds for metrics	Rejected — derive-only gate
Fork BOPA metrics inside FR-017	Rejected — reuse FR-015
Block Horizon 1B on FR-017	Rejected (M0 §9)
Change runtime to satisfy R-signals	Not needed — no blocker in M1